Arnav Arora

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EDUCATION

Fulton School of Engineering, Arizona State University **Tempe, Arizona**

Undergraduate Student at ***Barrett, The Honors College*** Bachelor's | Accelerated Master's Spring 2027

- **Major:,** Computer Science, Mathematical Statistics | **Minor:** Economics
- **GPA (CS | Statistics)*:*** 3.8/4.0 | **4**.0/4.0 | **Honors:** Dean's List, New American University Scholarship
- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Distributed Software Development, Machine/Deep Learning, Advanced Calculus, Probability & Statistics, Econometrics, Computer Networks. Object Oriented Programming. Cloud Computing. Statistical ML.

SKILLS

Programming Languages: Python, Java, C++, C#, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: React, Node.js, .NET, TensorFlow, PyTorch, CNNs, Transformers, Apache Spark (MLlib)

Cloud & DevOps: AWS, GitHub Actions, Terraform, API Gateway, AWS Lambda, Kafka, Docker

Databases & Data Tools: MongoDB, Power BI, CRM Systems

AI/ML & Research Tools: LLMs, Probabilistic Circuits, Statistical Models

Certifications & Training: AWS Machine Learning Engineer, Stanford Machine and Deep Learning Specializations.

Personal Interests: Indian Classical Music (Teacher and Performer)

WORK EXPERIENCE

Intel Corporation
Santa Clara – Remote

AI/ML Software Engineering Intern Feb 2025-Present

- Enhanced Intel's QIRI chatbot with hybrid retrieval (dense + sparse) and expert-matching LLMs, boosting answer speed by 80%.
- Built Holistic AI analytics platform using Python, Kafka, and MongoDB, reducing workforce reporting turnaround by 40%.
- Engineered containerized ETL microservices using Spark and serverless CI/CD (AWS Lambda, API Gateway), cutting deployment time by 54% and boosting analytics uptime.
- Built AI model marketplace for Intel Foundry with semiconductor domain-specific agents, code, and EDA tools—accelerated PLM workflows and knowledge reuse across chip design teams.

SCAI ML Labs ASU| Dr. YooJung Choi | NeurIPS Publication Summer 2025
Tempe, Arizona

ML Research Intern April 2024-Present

- Developed scalable pipelines for training probabilistic circuits (interpretable generative models) on multimodal biomedical datasets from Mayo clinic Hospital using ASU's high-performance SOL cluster.
- Designed algorithms for data preprocessing, reducing I/O and compute latency by 14% enabling faster experimentation on large-scale graphs.
- Conducted empirical evaluations using PyTorch/Julia, focusing on sample diversity, tractability and inference fidelity.

State-Street
Boston – Remote

Data Science Intern August 2024-December 2024

- Modernized legacy COBOL systems with Java and LLAMA AI, reducing inefficiencies by 33% and optimizing algorithmic performance.
- Built real-time AI dashboards in Power BI with Python and R, improving product service analysis and decision-making.
- Automated CI/CD pipelines with unit testing, boosting deployment speed and reliability by 30%.

Qualitest
Apex, North Carolina

SWE/AI-ML Intern January 2024 – April 2024

- Built RESTful APIs for a Quality Assurance framework, integrating React, Node.js, and MongoDB for seamless model communication.
- Engineered fault-tolerant database solutions using MongoDB and optimized queries for large-scale datasets.
- Debugged and Optimized feature selection using cloud-hosted machine learning models and integrated automated testing workflows.
- Applied object-oriented programming techniques in Java to develop reusable API modules, reducing code duplication.

Luminosity Lab
Tempe, Arizona

Undergraduate Technology Consultant August 2023-Present

- Partnered with **Apple** to co-design hardware-software integration strategies, aligning technical architectures with business goals.
- Optimized **Zoom's** virtual learning space architecture, reducing latency by 12% and driving a 20% increase in user engagement.
- Improved payment gateway reliability by 22% through streamlined scripting and architecture enhancements for high-traffic environments.

Leadership and Entrepreneurship

AI Orbitscape and LunarRights - https://orbitscape.space | https://lunarights.space

CTO & Co-Founder -Selected for Presentation at IAC 2025 (76th International Astronautical Congress) September 2024 – Present

Leading the next frontier of AI mission planning - scaling deep-tech infrastructure - showcasing globally at the 2025 IAC in Sydney, Australia.

- Shaped product strategy and secured three pilot partnerships for mission planning platform.
- Leading backend and API development using orbital mechanics libraries to power mission visualization tools for aerospace clients.

Technical Projects

Multimodal Medical Diagnosis Model | PyTorch, Probabilistic Circuits | Mayo Clinic Hospital, ASU

- Leading thesis research under Dr. YooJung Choi to develop a multimodal probabilistic AI model to predict disease progression.

AI Predictive Maintenance API | Python, Flask, ML Analysis | https://github.com/aarora80/predictMaintainence.git

- Built an API to predict Remaining Useful Life (RUL) using CMAPSS datasets, improving maintenance scheduling efficiency.

Distributed P2P Network Protocol | C, Sockets, Hash Maps | https://github.com/aarora80/SocketProject.git

- Built a high-performance P2P protocol in C with Hot Potato routing, custom DHT, and socket-based message passing across nodes.